

Week-3 Module

1. CODE -

```
#include <iostream>
#include <vector>
using namespace std;
void merge (vector<int> &arr, int left,
            int mid, int right) {
    int n1 = mid - left + 1;
    int n2 = right - mid;
    vector<int> L(n1), R(n2);
    for (int i=0; i<n1; i++)
        L[i] = arr[left+i];
    for (int j=0; j<n2; j++)
        R[j] = arr[mid+1+j];

    int i=0, j=0, k=left;
    while (i<n1 && j<n2) {
        if (L[i] <= R[j]) {
            arr[k] = L[i];
            i++;
        } else {
            arr[k] = R[j];
            j++;
        }
        k++;
    }
}
```

```

        k++;
    }
    while (i < n1) {
        arr[k] = L[i];
        i++;
        k++;
    }
    while (j < n2) {
        arr[k] = R[j];
        j++;
        k++;
    }
}

void mergeSort (vector<int> & arr,
                int left, int right) {
    if (left < right) {
        int mid = (left + right - left) / 2;
        mergeSort (arr, left, mid);
        mergeSort (arr, mid + 1, right);
        merge (arr, left, mid, right);
    }
}

int main() {
    ios_base::sync_with_stdio(false);
    cin.tie(NULL);
}

mergeSort (marks, 0, n-1);
for (int i = 0; i < n; i++) {

```

```

} cout << "\n";
return 0;
}

```

2. CODE -

```

#include <iostream>
using namespace std;
int arr[100005], temp[100005];
void merge(int l, int m, int r) {
    int i = l, j = m+1, k = l;
    while (i <= m && j <= r) {
        if (arr[i] <= arr[j]) temp[k++] = arr[i++];
        else temp[k++] = arr[j++];
    }
    while (i <= m) temp[k++] = arr[i++];
    while (j <= r) temp[k++] = arr[j++];
    for (i = l; i <= r; i++) arr[i] = temp[i];
}

void mergeSort(int l, int r) {
    if (l < r) {
        int m = l + (r - l) / 2;
        mergeSort(l, m);
        mergeSort(m+1, r);
        merge(l, m, r);
    }
}

```

```
int main() {
    int n;
    cin >> n;
    for (int i = 0; i < n; i++)
        cin >> arr[i];
    mergeSort(0, n-1);
    for (int i = 0; i < n; i++)
        cout << arr[i] << " ";
    cout << endl;
    return 0;
}
```

3. CODE -

```
#include <iostream>
using namespace std;
int arr[100005];
int partition(int low, int high) {
    int pivot = arr[high];
    int i = low - 1;
    for (int j = low; j <= high; j++)
        if (arr[j] <= pivot)
            swap(arr[i], arr[j]);
    swap(arr[i], arr[high]);
    return i;
}
```

```

int t = arr[i+1]; arr[i+1] =
    arr[high]; arr[high] = t;
return i+1;

```

```

}

```

```

void quicksort (int low, int high) {
    if (low < high) {
        int pi = partition (low, high);
        quicksort (low, pi-1);
        quicksort (pi+1, high);
    }
}

```

```

int main() {

```

```

    int n;

```

```

    cin >> n;

```

```

    for (int i=0; i<n; i++) cin >>
        arr[i];

```

```

    quicksort (0, n-1);

```

```

    for (int i=0; i<n; i++)

```

```

        cout << arr[i] << " ";

```

```

    cout << endl;

```

```

    return 0;

```

```

}

```

4.

CODE -

```

#include <iostream>
using namespace std;
long long arr[100005];
int partition (int low, int high) {
    long long pivot = arr[high];
    int i = low - 1;
    for (int j = low; j < high; j++) {
        if (arr[j] <= pivot) {
            i++;
            long long t = arr[i]; arr[i] =
                arr[j]; arr[j] = t; }
    }
    long long t = arr[i+1];
    arr[i+1] = arr[high];
    arr[high] = t;
    return i+1; }
void quickSort (int low, int high) {
    if (low < high) {
        int pi = partition (low, high);
        quickSort (low, pi-1);
        quickSort (pi+1, high); } }
int main() {
    int n;
    cin >> n;
    for (int i = 0; i < n; i++)

```

FileEditSelectionViewGoRunTerminalHelp

EXPLORER

SOURCE CODES

.vscode

StdRank.cpp

StdRank.exe

StdRank.cpp

```
1 #include <iostream>
2 using namespace std;
3
4 int arr[100005], temp[100005];
5
6 void merge(int l, int m, int r) {
7     int i = l, j = m + 1, k = l;
8     while (i <= m && j <= r) {
9         if (arr[i] <= arr[j]) temp[k++] = arr[i++];
10        else temp[k++] = arr[j++];
11    }
12    while (i <= m) temp[k++] = arr[i++];
13    while (j <= r) temp[k++] = arr[j++];
14    for (i = l; i <= r; i++) arr[i] = temp[i];
15 }
16
17 void mergeSort(int l, int r) {
18     if (l < r) {
19         int m = l + (r - l) / 2;
20         mergeSort(l, m);
21         mergeSort(m + 1, r);
22         merge(l, m, r);
23     }
24 }
25
26 int main() {
27     int n;
28     cout << "Enter the number of students: ";
29     cin >> n;
30
31     cout << "Enter the marks: ";
32     for (int i = 0; i < n; i++) {
33         cin >> arr[i];
```

CHAT

TERMINAL

PS C:\Users\hshar\Desktop\KCCITM TECH TRAINING MODULES\WEEEEK 3 MODULE\SOURCE CODES> cd "c:\Users\hshar\Desktop\KCCITM TECH TRAINING MODULES\WEEEEK 3 MODULE\SOURCE CODES\" ; if (\$?) { g++ StdRank.cpp -o StdRank } ; if (\$?) { .\StdRank }
Enter the number of students: 6
Enter the marks: 78 45 92 34 67 81
Sorted marks: 34 45 67 78 81 92
PS C:\Users\hshar\Desktop\KCCITM TECH TRAINING MODULES\WEEEEK 3 MODULE\SOURCE CODES>

OUTLINE

TIMELINE

JAVA PROJECTS

PROBLEMS

DEBUG CONSOLE

Filter (e.g. test, exclude...)

Java: Ready

Ln 12, Col 41 Spaces: 4 UTF-8 CRLF {} C++ Win32

28°C Cloudy

Search

Win32

17:51

08-07-2026

FileEditSelectionViewGoRunTerminalHelp

←→SOURCE CODES

↻

EXPLORER

SOURCE CODES

.vscode

EmpSal.cpp

EmpSal.exe

ProdPrice.cpp

ProdPrice.exe

StdRank.cpp

StdRank.exe

EmpSal.cpp

main()

1#include <iostream>

2using namespace std;

3

4int arr[100005], temp[100005];

5

6void merge(int l, int m, int r) {

7int i = l, j = m + 1, k = l;

8while (i <= m && j <= r) {

9if (arr[i] <= arr[j]) temp[k++] = arr[i++];

10else temp[k++] = arr[j++];

11}

12while (i <= m) temp[k++] = arr[i++];

13while (j <= r) temp[k++] = arr[j++];

14for (i = l; i <= r; i++) arr[i] = temp[i];

15}

16

17void mergeSort(int l, int r) {

18if (l < r) {

19int m = l + (r - l) / 2;

20mergeSort(l, m);

21mergeSort(m + 1, r);

22merge(l, m, r);

23}

24}

25

26int main() {

27int n;

28cout << "Enter the number of employees: ";

29cin >> n;

30

31cout << "Enter the salaries: ";

32for (int i = 0; i < n; i++) {

33cin >> arr[i];

CHAT

TERMINAL

PS C:\Users\hshar\Desktop\KCCITM TECH TRAINING MODULES\WEEK 3 MODULE\SOURCE CODES>

> cd "c:\Users\hshar\Desktop\KCCITM TECH TRAINING MODULES\WEEK 3 MODULE\SOURCE CODES\" ; if (\$?) { g++ EmpSal.cpp -o EmpSal } ; if (\$?) { .\EmpSal }

Enter the number of employees: 7

Enter the salaries: 45000 25000 70000 50000 30000 90000 60000

Sorted salaries: 25000 30000 45000 50000 60000 70000 90000

Lowest Salary: 25000

Highest Salary: 90000

Difference: 65000

PS C:\Users\hshar\Desktop\KCCITM TECH TRAINING MODULES\WEEK 3 MODULE\SOURCE CODES>

OUTLINE

TIMELINE

JAVA PROJECTS

PROBLEMS

DEBUG CONSOLE

Filter (e.g. test, exclude...)

28°C

Cloudy

Search

Win32

17:58

08-07-2026

File Edit Selection View Go Run Terminal Help

EXPLORER

SOURCE CODES

.vscode

Bookids.cpp

Bookids.exe

EmpSal.cpp

EmpSal.exe

ProdPrice.cpp

ProdPrice.exe

RaceTime.cpp

RaceTime.exe

StdRank.cpp

StdRank.exe

OUTLINE

TIMELINE

JAVA PROJECTS

Bookids.cpp

```
1 #include <iostream>
2 using namespace std;
3
4 long long arr[100005]; // long long matches the ID constraint
5
6 int partition(int low, int high) {
7     long long pivot = arr[high];
8     int i = low - 1;
9     for (int j = low; j < high; j++) {
10         if (arr[j] <= pivot) {
11             i++;
12             long long t = arr[i]; arr[i] = arr[j]; arr[j] = t;
13         }
14     }
15     long long t = arr[i + 1]; arr[i + 1] = arr[high]; arr[high] = t;
16     return i + 1;
17 }
18
19 void quickSort(int low, int high) {
20     if (low < high) {
21         int pi = partition(low, high);
22         quickSort(low, pi - 1);
23         quickSort(pi + 1, high);
24     }
25 }
26
27 int main() {
28     int n;
29     cout << "Enter the number of books: ";
30     cin >> n;
31
32     cout << "Enter the book IDs: ";
33     for (int i = 0; i < n; i++) {
```

CHAT

TERMINAL

PS C:\Users\hshar\Desktop\KCCITM TECH TRAINING MODULES\WEEEEK 3 MODULE\SOURCE CODES>

> cd "c:\Users\hshar\Desktop\KCCITM TECH TRAINING MODULES\WEEEEK 3 MODULE\SOURCE CODES\" ; if (\$?) { g++ Bookids.cpp -o Bookids } ; if (\$?) { .\Bookids }

Enter the number of books: 5

Enter the book IDs: 1025 1001 1050 1010 1005

Sorted IDs: 1001 1005 1010 1025 1050

PS C:\Users\hshar\Desktop\KCCITM TECH TRAINING MODULES\WEEEEK 3 MODULE\SOURCE CODES>

PROBLEMS

DEBUG CONSOLE

Filter (e.g. text, exclude...)

28°C Cloudy

Search

Win32

18:05 08-07-2026

File Edit Selection View Go Run Terminal Help

EXPLORER

SOURCE CODES

.vscode

Bookids.cpp

Bookids.exe

EmpSal.cpp

EmpSal.exe

ProdPrice.cpp

ProdPrice.exe

RaceTime.cpp

RaceTime.exe

StdRank.cpp

StdRank.exe

StoreSales.cpp

StoreSales.exe

StoreSales.cpp

```
1 #include <iostream>
2 using namespace std;
3
4 int arr[100005];
5
6 int partition(int low, int high) {
7     int pivot = arr[high];
8     int i = low - 1;
9     for (int j = low; j < high; j++) {
10         if (arr[j] >= pivot) { // '>=' changes it to descending order
11             i++;
12             int t = arr[i]; arr[i] = arr[j]; arr[j] = t;
13         }
14     }
15     int t = arr[i + 1]; arr[i + 1] = arr[high]; arr[high] = t;
16     return i + 1;
17 }
18
19 void quickSort(int low, int high) {
20     if (low < high) {
21         int pi = partition(low, high);
22         quickSort(low, pi - 1);
23         quickSort(pi + 1, high);
24     }
25 }
26
27 int main() {
28     int n;
29     cout << "Enter the number of sellers: ";
30     cin >> n;
31
32     cout << "Enter the sales values: ";
33     for (int i = 0; i < n; i++) {
```

CHAT

TERMINAL

PS C:\Users\hshar\Desktop\KCCITM TECH TRAINING MODULES\WEEEEK 3 MODULE\SOURCE CODES>

> cd "c:\Users\hshar\Desktop\KCCITM TECH TRAINING MODULES\WEEEEK 3 MODULE\SOURCE CODES\" ; if (\$?) { g++ StoreSales.cpp -o StoreSales } ; if (\$?) { .\StoreSales }

Enter the number of sellers: 6

Enter the sales values: 1200 5000 3000 2500 7000 4500

Sorted sales figures: 7000 5000 4500 3000 2500 1200

Top 3: 7000 5000 4500

Average: 5500

PS C:\Users\hshar\Desktop\KCCITM TECH TRAINING MODULES\WEEEEK 3 MODULE\SOURCE CODES>

OUTLINE

TIMELINE

JAVA PROJECTS

PROBLEMS

DEBUG CONSOLE

Filter (e.g. test, testcode...)

28°C Cloudy

Search

Ln 50, Col 2 Spaces: 4 UTF-8 CRLF C++ Win32

18:07 08-07-2026